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Buried-Contact Organic Field-Effect Transistor: The Way of Alleviating Drawbacks from Interfacial Charge Transfer

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Facile charge transfer between source/drain (S/D) electrodes and organic semiconductor (OSC) channel is crucial for high-mobility organic field-effect transistors (OFETs). Herein, a novel OFET geometry is developed by modifying a top-contact bottom-gate device structure, termed a buried-contact OFET, enabling close proximity between the S/D-OSC interface and conducting channel, consequently decreasing the access contact resistance ($R_{C,acc}$) and overall contact resistance (R_C). Conventional post-thermal annealing is combined with a burying pressure (pressure-thermal annealing (PTA)). The synergistic effect of thermal and pressure annealings leads to the softened OSC layer enabling metal electrodes to bury inward by applied pressure. This process induces structural transitions from a top-contact to buried-contact configuration, as verified by atomic force microscopy and finite element simulations. Transfer line method and 4-probe measurements revealed that PTA reduces the contact by 1/3 (65 kΩ cm) and the source-to-drain voltage waste due to charge injection from 52% to 31%. Consequently, the field-effect mobility is four times higher than that of a conventional thermally annealed top-contact OFET. The density of deep traps (N_{tr}) is mainly distributed in the OSC bulk responsible for charge injection. Remarkably, the N_{tr} decreased 30-fold using PTA, resulting in a shallow sub-threshold region and a threshold voltage close to zero.

1. Introduction

Organic field-effect transistors (OFETs) have been intensively studied over the past few decades because of their potential for cost-effective large-scale manufacturing via solution processes.[1–6] Their capability to be fabricated not only on hard substrates[4,5] but also on platforms such as flexible polymers[6,7] and paper,[8–10] owing to their low processing temperatures, suggests their potential for expansion into applications such as flexible displays,[10,11] sensors,[12,13] and electronic skin.[14,15] However, the persistent issue of low electrical performance and reliability remains a barrier to the practical application of OFETs. One of the main problems is the inherent charge transfer across the various interfaces present within the device.[16–19] In particular, the source (S)-to-organic semiconductor (OSC) and OSC-to-drain (D) charge-transfer interfaces play important roles in determining the electrical properties of OFETs. Therefore, the realization of high-performance OFETs

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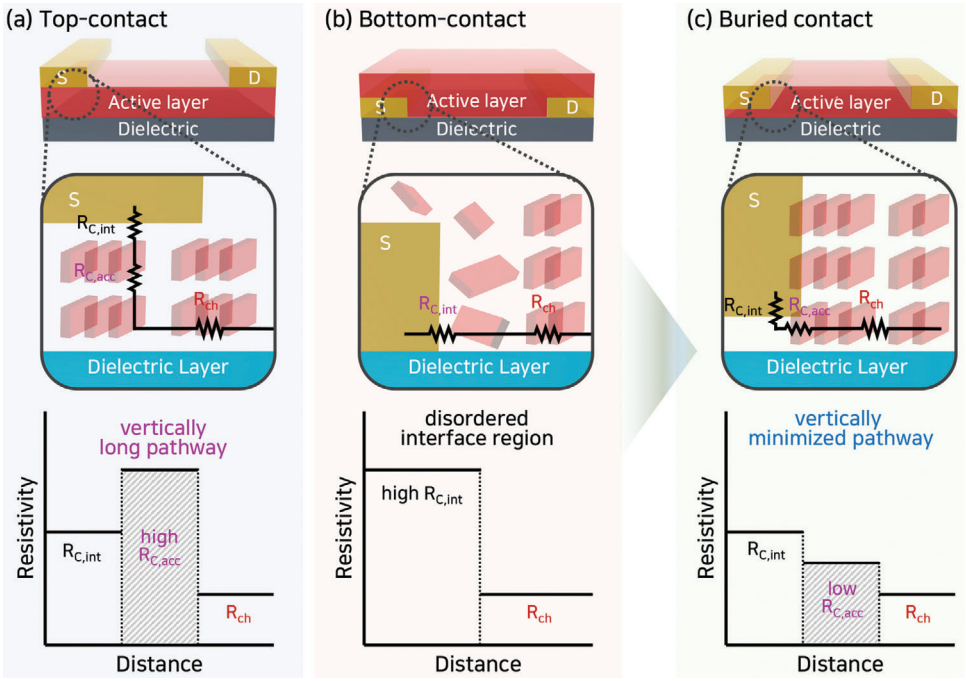


Figure 1. Schematic of the crystalline arrangements within the OSC film and changes in the expected resistances resulting from the arrangements of the representative OFET structures, a) top-contact, b) bottom-contact, and c) buried-contact structures.

following Shockley behavior relies on effective control of the contact resistance (R_C) arising at the S/D-to-OSC interface.[2,20–24]

One of the fundamental reasons for the high R_C is the Schottky barrier that arises owing to the energy level mismatch between the OSC and metal S/D. In previous studies, the energy-level mismatch at the Schottky barrier interface was controlled by applying various strategies, including introducing a doped interface,[25,26] inserting a metal oxide interfacial layer,[27,28] and forming self-assembled monolayers.[29–32] However, recent studies on the origin of R_C have revealed that R_C is composed of the interfacial contact resistance ($R_{C,int}$) and access contact resistance ($R_{C,acc}$), and $R_{C,acc}$ has a greater contribution to R_C than $R_{C,int}$[33–35] $R_{C,int}$ arises from to the substantial energy-level mismatch between the OSC and S/D, corresponding to the Schottky barrier. More importantly, $R_{C,acc}$ is caused by charge scattering/trapping in the pathway of the injected charge carriers across the OSC bulk to reach the accumulated channel. A lower R_C resulting from a lower $R_{C,acc}$ was achieved by reducing the thickness of the OSC film to shorten the charge injection pathway across the OSC bulk.[34–37] However, the ultrathin OSC layer can reduce the resistance of the OFET to oxygen and moisture and the uniformity of the OSC layer, leading to inferior performance reliability.[38,39] Furthermore, there is a risk that the balance between the intrinsic tensile and compressive strains present among OSC molecules cannot be maintained, leading to the potential for easy formation of defects in the OSC crystalline structures over time.[37,40] Thus, despite the emphasized need to reduce $R_{C,acc}$, an effective system design to address this issue has not been achieved.

In this study, we propose an OFET with a buried-contact structure, formed by modifying the top-contact structure, where S and D are embedded into the OSC layer, as a unique strategy for shortening the charge-transport pathway from the OSC/electrode interface to the channel region, thereby reducing the $R_{C,acc}$ and the overall R_C. This approach addresses the shortcomings of the methods discussed above for reducing the thickness of the organic active layer to decrease $R_{C,acc}$. By effectively shortening the pathway through which charge carriers are transported, we aim to achieve a reduction in R_C and an overall improvement in the device performance through this new approach. To fabricate the OFET device with the buried-contact structure, a pressure-thermal annealing process (PTA) is applied, in which thermal annealing is combined with concurrent pressure application. This technique successfully embeds the S/D electrodes, initially formed as a top structure over the active layer, into the active layer. The field-effect mobility (μ_{FET}) of the buried-contact OFETs is increased by over tenfold and the trap density (N_{tr}) and R_C are reduced by approximately threefold compared to those of the pristine OFETs. By comparing the voltage drop and μ_{FET} characteristics of various reference groups obtained through both 4-probe and 2-probe measurements, the effects of *R*s in the organic active layer and at the interface on the driving device performance are separately analyzed. The results prove the impact and utility of the buried-contact OFET structure in enhancing the performance of OFET devices based on systematic analysis.

2. Results and Discussion

Figure 1 illustrates the representative OFET structures with the top and bottom contacts, alongside the developed buried-contact structure. The top-contact structure in which the S/D electrodes were patterned onto the OSC thin film via thermal evaporation exhibited a relatively low $R_{C,int}$. However, prior to lateral charge transport along the channel, which is mainly on the intermolecular π-stacking direction, the charge carriers injected from

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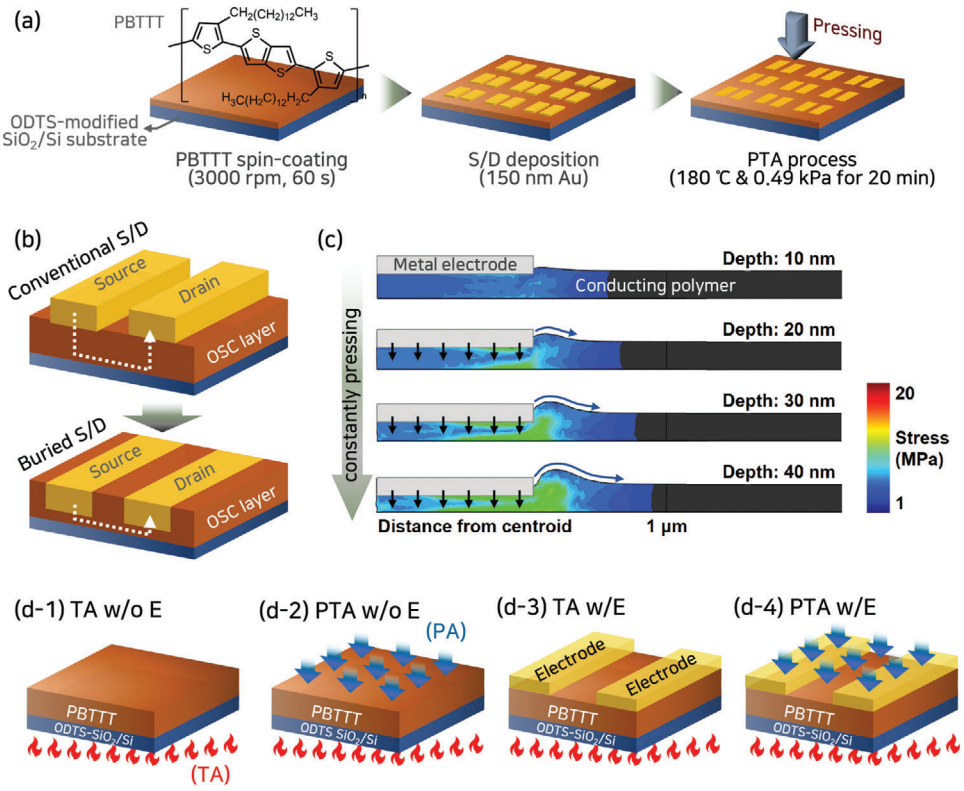


Figure 2. Schematic summarizing a) fabrication of buried-contact OFET and b) difference between top-contact and buried contact structure. c) Finite element simulations by applying the von Mises stress with different indentation depths. Dark gray on the substrate indicates region where the magnitude of von Mises stress is less than 1 MPa. d) Schematic of OFET groups designed for systematic comparison of the effectiveness of the buried-contact structure.

electrode should travel across the OSC bulk in the interlamellar stacking direction. Consequently, $R_{C,acc}$ contributed more to R_C than $R_{C,int}$. Modification of the OSC-S/D interface in a top-contact device is difficult in reality, posing limitations in introducing an interlayer to reduce the R_C.[3,41] On the other hand, in the bottom-contact structure, the OSC molecules deposited on the S/D electrode and dielectric layer undergo different surface-mediated crystallization processes. This induces a high $R_{C,int}$ owing to the inhomogeneous molecular orientation on the different surfaces. The proposed buried-contact OFET combines the advantages of both structures. By burying the electrodes within the organic active layer, the low $R_{C,int}$ of top-contact structures can be combined with the benefit of the lower $R_{C,acc}$, which stems from the shorter pathway from the electrode/active layer interface to the channel region.

Figure 2a illustrates the process for fabricating the buried-contact OFET. The organic active layer, PBTTT-C14 (hereafter referred to as PBTTT), was spin-coated onto the substrate, followed by deposition of the S/D electrodes. Post-annealing was conducted at 180 °C for 20 min while applying a pressure of 0.49 kPa. This post-annealing method is henceforth termed PTA. Owing to the fluidity of the PBTTT layer imparted by thermal energy, the PBTTT film allowed the metal electrodes to penetrate inward when subjected to an applied pressure, inducing structural transition from a top-contact configuration to a buried-contact configuration, as depicted schematically in Figure 2b. The burial of the electrodes into the organic active layer during the PTA process was validated through finite element simulations by applying the magnitude of the von Mises stress in Figure 2c; the calculation method is described in detail in the Supporting Information.

Figure 2d presents a comparison of various groups designed to systematically verify the effectiveness of the buried-contact structure. The top-contact OFET in which the electrode was formed only after thermal annealing of the organic semiconductor layer was named TA w/o E. The device in which the electrode was formed after applying thermal and pressure annealing simultaneously was named PTA w/o E. The devices in which thermal annealing and pressure+thermal annealing were performed after formation of the electrode on the organic semiconductor layer are denoted as TA w/E and PTA w/E, respectively. The TA w/o E sample is designed to assess the impact of the improved crystalline structure on the device performance by performing only TA on the OSC layer, whereas the PTA w/o E sample enables evaluation of the effects of the additional pressure applied during the TA process on the OSC. When post-annealing is performed with the electrode formed on the OSC layer, there is the potential for a different crystallization process at the interface with the electrode; this was examined using the TA w/E device. Finally, the buried-contact structure was formed as follows: when pressure was applied to the electrode on the OSC layer, which was fluidized by thermal energy, the electrode was buried inside, forming the

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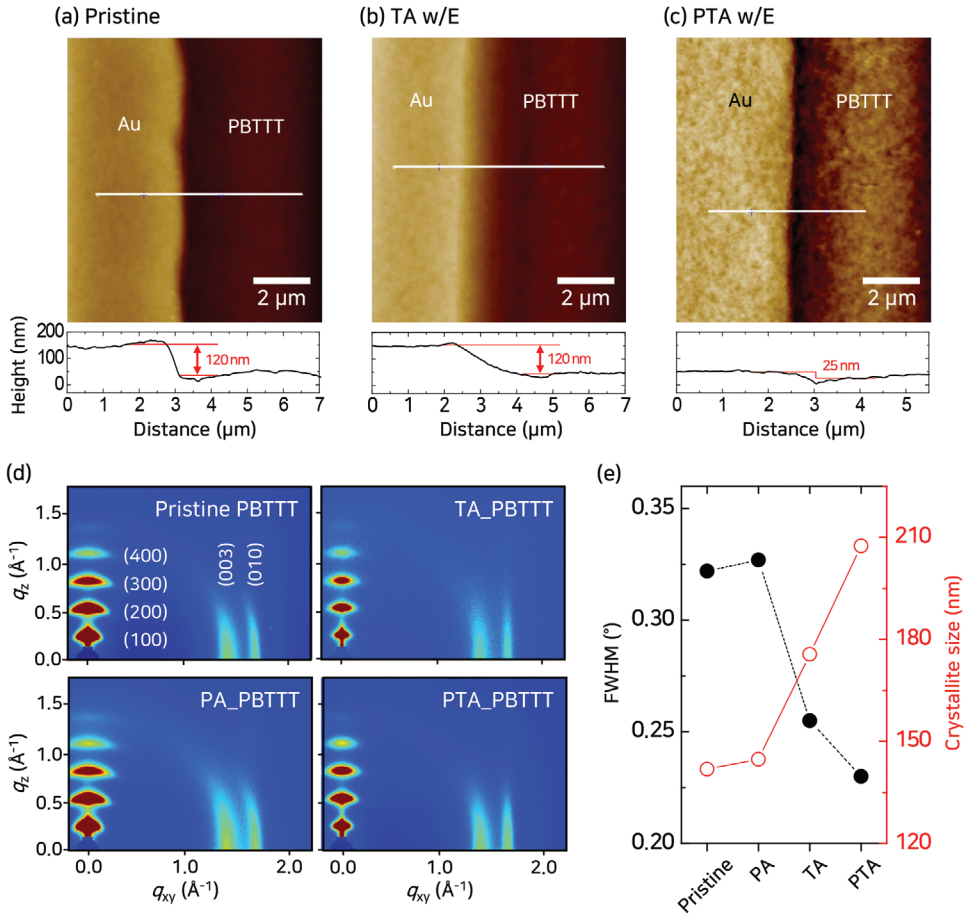


Figure 3. Surface morphologies and corresponding height profiles obtained through AFM analysis of the boundary sections where the electrode and OSC layers meet for the a) pristine, b) TA w/E, and c) PTA w/E devices. d) 2D GIXD data for PBTTT films after various post-annealing processes. e) FWHM and the crystallite size determined from signal of (200) peak in the q_z direction by using the results in (d).

desired structure. By preparing various samples, the effect of the buried-contact structure and other post-annealing processes on the performance of the OFET device could be systematically compared, striving to ensure the reliability of the experimental results.

To verify whether the electrodes are truly buried inside the PBTTT semiconducting layer by applying the PTA process, the atomic force microscopy (AFM) data for the boundary of the electrode and OSC are shown in **Figure** 3a–c. Figure 3a shows the morphology of the boundary for the pristine device without any post-processing. The boundary of the electrode can be clearly observed, where the thickness was 150 nm, as measured by using a vacuum evaporator monitor; the electrode was formed almost completely accurately on the PBTTT layer. For the TA w/E device (Figure 3b), although the boundary was less defined owing to the difference in the thermal dynamic motion supplied to the PBTTT polymer and the electrode metal, the height of the electrode did not differ significantly from the originally formed height. However, for the PTA w/E device (Figure 3c), the electrode seemed to be almost buried in the PBTTT layer, primarily because of the intense pressure applied to the protruding electrode. This morphological analysis confirms that the PTA process can successfully form OFET devices with the buried-contact structure.

Figure 3d shows the 2D grazing incident X-ray diffraction (2D GIXD) data, highlighting how the crystalline structure of the PBTTT organic active layer changed during the TA, PA, and PTA processes. For the pristine PBTTT film, the (100), (200), and (300) peaks appeared at positions of 0.29, 0.58, and 0.85 Å^{-1} in the q_z direction, respectively, and the (003) peak appeared at 1.43 Å^{-1} in the q_{xy} direction. These results suggest that the crystalline structure within the PBTTT film is multi-stacked in a direction perpendicular to the surface, indicating that the conjugated structures constituting the molecular backbone within the crystalline structures are aligned perpendicular to the surface. Moreover, the peaks identified in the q_{xy} and q_z directions for the pristine PBTTT film, which were reflected along the Debye ring, were significantly narrower than those of the films subjected to TA. This indicates that the orientation and size of the crystalline domains in the PBTTT film were improved by the TA process, which ultimately improved the crystallization quality of the film. Conversely, the crystalline characteristics of the film formed by the PA process, in which only pressure was applied to the PBTTT film, were not significantly different from those of the pristine film. This reveals that the PA process, which does not induce dynamic thermal processes in the OSC molecules, does not contribute to improving the crystalline structure through molecular

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Table 1. Summary of PBTTT FET parameters for various post-annealed devices.

	Pristine	TA w/o E	PTA w/o E	TA w/E	PTA w/E
μ_{FET} ($\times 10^{-1}$ cm^2 V^{-1}s^{-1})	0.16 ± 0.06	0.52 ± 0.11	0.58 ± 0.07	1.40 ± 0.16	1.80 ± 0.09
V_{th} (V)	12.9 ± 4.1	9.4 ± 4.7	7.8 ± 2.3	8.5 ± 2.7	5.4 ± 1.6
N_{tr} (cm^{-2})	9.1×10^{11}	4.6×10^{11}	6.0×10^{11}	2.6×10^{12}	2.7×10^{10}
SS (V dec^{-1})	31.3 ± 7.8	20.6 ± 5.2	18.4 ± 5.1	13.7 ± 6.9	3.2 ± 2.4

rearrangement in the film, unlike the TA process. The reflection patterns of the PTA_PBTTT film subjected to the proposed PTA process were not significantly different from those of the TA film; however, an evident increase in the intensities was observed. These findings suggest that the PA process, which had no significant effect on the molecular crystalline rearrangement, contributed to improving the film crystallinity when applied concurrently with thermal energy. Figure S2 (Supporting Information) presents the 1D plots in the out-of- and in-plane directions for the 2D GIXD data shown in Figure 3d; the full-width-at-half-maximum (FWHM) of the (200) peak obtained in the out-of-plane direction was used to calculate the crystallite size using the Scherrer equation[42–44] (see Figure 3e). For the pristine PBTTT film and the film formed via the PA process, the average diameters of the crystallites were ≈142 and 145 nm, respectively, consistent with the previous descriptions. On the other hand, for the PBTTT films subjected to the TA and PTA processes, the diameter of the crystallites increased drastically to 175 and 208 nm, respectively. Although the results shown in Figure 3e may not be highly reliable, they still offer valuable insights as reference data reflecting the enhancement of the crystalline qualities within the organic active layers, depending on the process.

To verify the effectiveness of the proposed buried-contact structure (formed through the PTA process) in improving the OFET performance, a control group was prepared using various post-processing methods, and the performance of the devices was comparatively analyzed. Figure S5 (Supporting Information) shows the representative output and transfer characteristics of the OFET devices prepared in diverse manners; the average performance parameters of more than ten individually prepared devices are summarized in **Table 1**. The improved performance of the buried-contact OFET device was verified by analyzing μ_{FET}, which is considered the most crucial parameter for assessing the performance of OFETs; the results are presented in **Figure 4a**. The highest μ_{FET} was obtained with the PTA w/E device having a buried-contact structure, and this outcome remained consistent even when the channel length was varied from 30 to 100 µm. The performance of the five OFET devices compared in this study showed some discrepancies when the channel length was 100 µm. However, because the performance variation followed similar trends with channel lengths of 30 and 50 µm, the μ_{FET} and threshold voltage (V_{th}) were reorganized to more clearly show the performance trend depending on the device type by focusing on the channel length of 30 µm (see Figure 4b; Figure S6, Supporting Information).

As shown in Figure 4b and Table 1, the μ_{FET} of the pristine device was 0.016 ± 0.006 cm^2 V^{-1} s^{-1}. For the TA w/o E device in which the PBTTT layer was thermally annealed at 180 °C, the μ_{FET} increased by ~3.5 times (to 0.052 ± 0.011 cm^2 V s^{-1}). For the PTA w/o E device in which the PBTTT layer was simultaneously subjected to the PA and TA processes, the μ_{FET} was 0.058 ± 0.007 cm^2 V s^{-1}, which is not significantly different from that of the TA w/o E device. This result can be explained by confirming that simply applying pressure to the film does not affect the reorientation and crystallization characteristics of the OSC molecules, as indicated by the 2D GIXD results in Figure 3c and the AFM morphology analysis in Figure S3 (Supporting Information) and Figure 4. On the other hand, for the TA w/E and PTA w/E devices where post-processing was performed on the OFET devices in which both the electrode and OSC layers were formed, a notable improvement in the μ_{FET} was confirmed, with respective values of 0.14 ± 0.016 and 0.18 ± 0.009 cm^2 V·s^{-1}. Current studies aiming at boosting the performance of OFET devices through thermal annealing have predominantly focused on applying thermal energy to the OSC layer to optimize the molecular orientations and crystalline structures. However, this study reveals that thermal annealing of both the electrode and OSC layer after electrode formation can contribute more significantly to enhancing the performance of the devices. Moreover, the elevated μ_{FET} value observed in the case of PTA w/E compared to TA w/E suggests that the effectiveness of pressure annealing is enhanced when accompanied by thermal annealing. This synergistic variation of μ_{FET} closely follows the trend observed for V_{th}. Specifically, the OFET device with the PTA w/E setup displayed an ideal device driving performance, converging closest to 0 V. In other words, the innovative buried-contact structure is effective for optimizing the performance of OFET devices.

To understand the variation of μ_{FET} owing to the various additional post-processes applied to the OFET device on R_C, the $R_C W$ was extracted using the transfer line method (TLM), as shown in Figure 4c[45–47] (here, W denotes the channel width). When $V_G = -10$ V, the $R_C W$ was the highest for the pristine device, and decreased in the following order: TA w/E, TA w/o E, PTA w/o E, and PTA w/E. As V_G increased to −60 V, a slight inversion in $R_C W$ was observed for the TA w/o E and PTA w/o E devices, but the difference was negligible. This variation arises because the enhanced crystallinity of the OSC layer led to a reduction in R_C.[36,48] Note that for the PTA w/E device having the buried-contact OFET structure, the R_C was smaller than that of the other devices. As previously mentioned, applying pressure to the OSC layer during post-processing did not dominantly influence the crystalline structures and surface morphologies of the films or the OFET device performance, which can be discerned by directly comparing the TA w/o E and PTA w/o E devices. The TA w/E device had the second highest $R_C W$ value, which may be due to the differences in the thermal energy transfer characteristics between the electrode and the OSC layer, leading to variations in the rate and distribution of thermal energy across these two regions.

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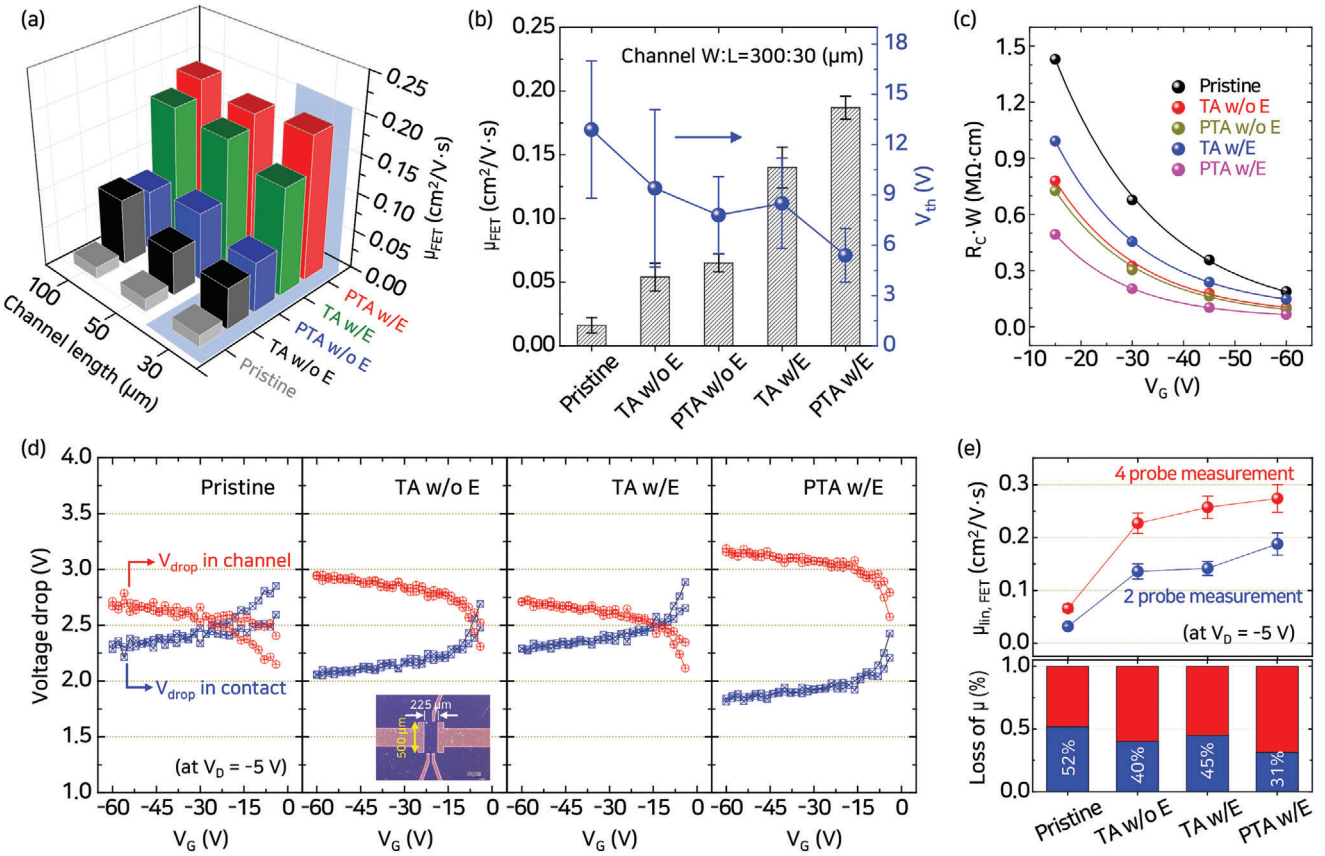


Figure 4. a) Change in μ_{FET} of PBTTT FET devices according to the various post-annealing processes with variation of channel length from 30 to 100 μm. b) Change in μ_{FET} and V_{th} for PBTTT FET devices with channel length of 30 μm subjected to various post-annealing processes. c) Comparison of R_C data for various devices measured through TLM according to V_G variation. d) Comparison of results of decoupling the voltage drops originating in the channel and contact region based on the 4-probe measurement. e) $\mu_{lin,FET}$ obtained from the 4-probe and 2-probe measurements (top) and loss of $\mu_{lin,FET}$ by the R_C (bottom) for various devices. Data were obtained at V_D and V_G of −5 and −60 V, respectively.

In addition, the higher-than-expected $R_C W$ for the TA w/E device is one of the reasons why the changes in the μ_{FET} were not clearly elucidated earlier. Interpreting these results holistically, the OFET device with the buried-contact structure formed by the PTA w/E process is highly likely to have a higher $R_{C,int}$, similar to the TA w/E device. However, owing to the drastic reduction in the distance between the electrode and the channel, a significant decrease in $R_{C,acc}$ is expected. Consequently, the overall R_C, which is the sum of $R_{C,int}$ and $R_{C,acc}$, is anticipated to increase or decrease significantly compared to that of the other devices.

The variations observed in the TLM were also consistent when using the 4-probe measurement, which was employed to extract the electrical properties without the interference of R_C, as depicted in Figure 4d–e. After acquiring the transfer curves in the linear regime, the voltage drops that occurred between the S and D electrodes (V_{SD}) were decoupled from those in the channel ($V_{channel}$) and contact ($V_{contact}$), as shown in Figure 4d. A higher $V_{contact}$ corresponds to a higher R_C, and the $V_{contact} - V_G$ relationship shown in Figure 4d is consistent with the TLM trends shown in Figure 4c. More importantly, $V_{contact}$ was significantly lower for the PTA w/E device than for the other devices. For the PTA w/E device, $V_{contact}$ was significantly lower than $V_{channel}$, indicating that the applied V_{SD} was more effectively consumed in the channel region, thereby allowing more efficient current flow in the channel.

The μ_{FET}s in the linear regime ($\mu_{lin,FET}$) excluding R_C, were also extracted from the 4-probe measurement (Figures 4e; Figure S8, Supporting Information).[49] As shown in the upper panel of Figure 4e, the $\mu_{lin,FET}$ values were higher in the 4-probe measurement than in the 2-probe measurement, because the inherent charge transfer outweighed R_C, which was excluded in the 4-probe measurement. Thus, by comparatively evaluating the ratios of $\mu_{lin,FET}$ from these two measurements, the R_C-induced loss of $\mu_{lin,FET}$ can be determined as: loss of $\mu = 1 - \mu_{lin,FET}{}^{2-probe}/\mu_{lin,FET}{}^{4-probe}$. As shown in the bottom panel of Figure 4e, the loss of μ was 52% for the pristine device, but decreased to 31% for the PTA w/E device. The difference in $\mu_{lin,FET}$ from the 4-probe and 2-probe measurements was lowest for the PTA w/E device. Note that the $\mu_{lin,FET}{}^{4-probe}$ was similar for the TA w/o E, TA w/E, and PTA w/E devices, whereas $\mu_{lin,FET}{}^{2-probe}$ was significantly higher for the PTA w/E device. This implies that the R_C was efficiently reduced in the developed buried-contact OFET, consequently enhancing charge transport.

In addition to the charge carrier mobility, the developed process affected other device properties. In particular, there were significant changes in the threshold voltage (V_{th}) and onset voltage

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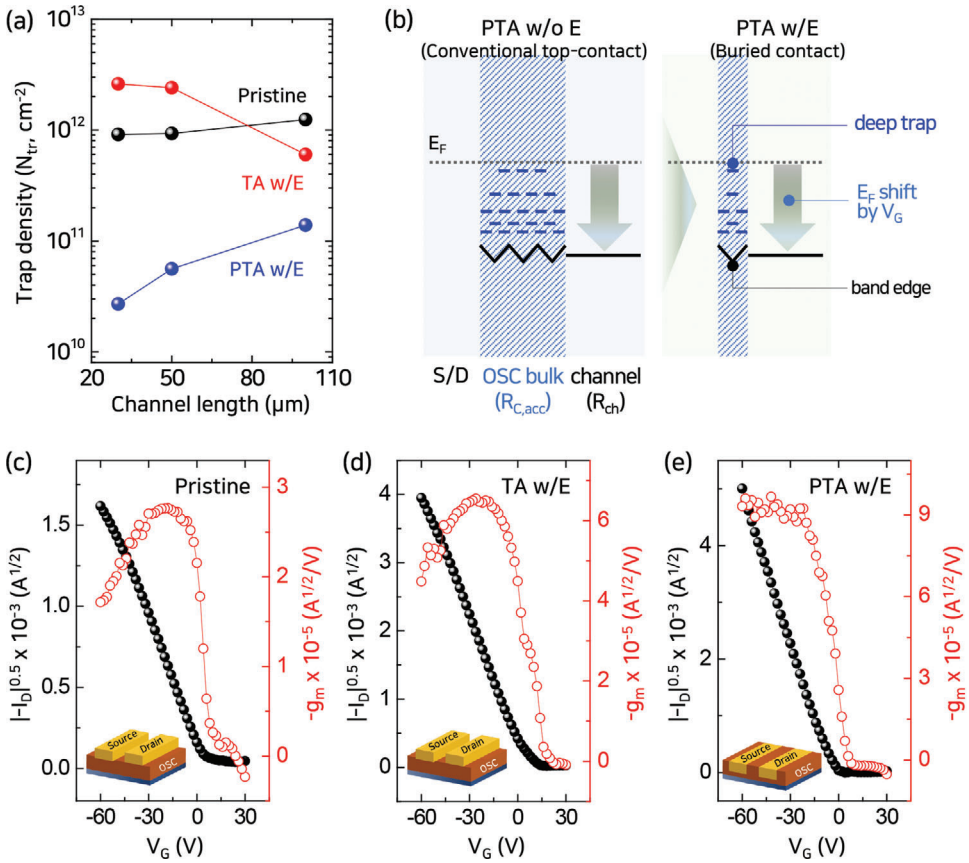


Figure 5. a) Results of N_{tr} calculation for PBTTT FET devices according to channel length for the pristine, TA w/E, and PTA w/E devices. b) Origin of N_{tr} decrease in PTA w/E. c–e) g_m analysis of PBTTT FET devices for the pristine, TA w/E, and PTA w/E devices.

(V_{onset}) of the various devices (Table 1; Figure S5, Supporting Information). It is known that the gap between V_{th} and V_{onset} is increased by the presence of deep traps, whereas the charge carrier mobility can be reduced by the presence of shallow traps.[50,51] The following equation was used to quantify the density of deep traps (N_{tr}): $N_{tr} = C_i|V_{onset} - V_{th}|/q$, where C_i is the areal capacitance of the employed dielectric layer, and the q is the elementary charge.[50,52,53] As summarized in Table 1 and **Figure 5a**, N_{tr} was 20–100 times lower in the developed buried-contact OFET, indicating an extremely low density of deep traps. Notably, N_{tr} was significantly lower in the PTA w/E (2.7×10^{10} cm^{-2}) than in the PTA w/o E (6.0×10^{11} cm^{-2}) device, suggesting that the deep traps were mainly distributed in the charge injection area. It is noted that the N_{tr} is larger in the TA w/E case than in the pristine case for channel lengths of 30 and 50 μm. As mentioned earlier, this result is due to the difference in thermal energy transfer characteristics between the electrode and the OSC layer in the TA w/E case. This is expected to be because the discontinuity of charge transfer increases at the interface between the two. This discontinuity at the interface significantly affects the FET device with the short channel lengths of 30 and 50 μm. On the other hand, in the case of a 100 μm channel length where the distance between electrodes is relatively long, the effect from the bulk is more pronounced than the interfacial characteristics, so the N_{tr} could be expected to be reduced compared to the pristine case. Figure 5b shows a schematic explaining the significant reduction in the deep traps in the developed device. In the area of the OSC bulk allowing vertical charge transport between the S/D and OSC channels, which is known to affect $R_{C,acc}$, the region between the lamellar stacks, for which the distance is influenced by the non-conducting alkyl side chains, presumably contains a significant portion of deep traps between the Fermi level and the HOMO level of PBTTT. In contrast to the PTA w/o E, this area rich in deep traps can be physically reduced in the PTA w/E because the S/D electrodes push down on the OSC layer.

To confirm the variation in the V_G-dependent μ_{FET}, which is one of the primary device parameters influenced by the increased R_C, the transconductance (g_m) of the representative devices (pristine, TA w/E, and PTA w/E cases) was comparatively analyzed based on the transfer characteristics, as illustrated in Figure 5c–e. Consistent with earlier discussions regarding R_C and N_{tr}, the g_m curves for both the pristine and TA w/E devices changed, even in the V_G section of the on-state regime, whereas the g_m values for the PTA w/E device appeared almost constant. This difference arises because the OFET devices with the buried-contact structure have much lower R_C and N_{tr} values, proving the effectiveness of the designed OFET structure.

3. Conclusion

This study introduces a novel strategy for lowering the R_C in OFET devices by forming a buried-contact structure via the PTA

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w/E process, combined with the benefits of reducing the $R_{C,acc}$, which arises because of charge scattering or trapping in the pathway as the injected charge moves from the OSC bulk to the channel region. The μ_{FET}, one of the most important performance parameters for OFETs, was compared, demonstrating that the PTA w/E OFET device with the buried-contact structure had a μ_{FET} of 0.18 ± 0.009 $cm^2\ V^{-1}s^{-1}$, compared to 0.016 ± 0.006, 0.052 ± 0.011, 0.058 ± 0.007, and 0.14 ± 0.016 $cm^2\ V^{-1}s^{-1}$ for the pristine, TA w/o E, PTA w/o E, and TA w/E devices, respectively, which were prepared for comparison. Notably, for the PTA w/E device, the μ_{FET} increased by up to 3.5 and 1.5 times, respectively, compared to that of the PTA w/o E device subjected to the same post-annealing process without forming the buried-contact structure and the TA w/E device, which only underwent thermal annealing. Moreover, the OFET device subjected to the PTA w/E process demonstrated the optimal device performance, converging closest to 0 V for V_{th}, operating quickly with a low SS, and showing a dramatic 45-fold reduction of N_{tr} versus that of the pristine device. Finally, to validate the objective of this study, which was to reduce the R_C of the OFET device by introducing a buried-contact structure, a comparative analysis using TLM and 4-/2-probe measurements was conducted, confirming that the device performance was significantly improved owing to the efficient mitigation of the actual R_{Cacc}. In essence, the buried-contact OFET structure is a pivotal breakthrough in achieving optimal polymer-based FET device performance by substantially lowering the R_C.

4. Experimental Section

Sample Preparation: Heavily n-doped Si wafers with 300 nm thick thermally grown oxide (SiO_2) were used as the substrate, gate electrode, and gate dielectric. The SiO_2/Si wafer was immersed in piranha solution for 20 min at 180 °C. After cooling for 50 min, the sample was washed with a large amount of distilled water and dried with nitrogen. The residual moisture on the Si/SiO_2 wafer was removed by applying a vacuum in the reactor for 1 hr, and the cleaned Si/SiO_2 wafer was reacted with a 10 mM octadecyltrichlorosilane (ODTS, Sigma–Aldrich) solution in anhydrous toluene (Sigma–Aldrich) under argon atmosphere for 2 hrs. The modified substrate was then lightly washed with toluene, dried with nitrogen, baked in a 120 °C oven for 20 min to complete the reaction for placing ODTS dangling bonds on the surface, and finally washed with toluene, acetone, and ethanol.

OFET Device Fabrication: PBTTT was dissolved in anhydrous toluene (Sigma–Aldrich) to a concentration of 1 wt%. The solution was stirred for 1 hr at 80 °C and directly spin-coated (3000 rpm, 60 s) on the ODTS-modified substrate to from an ≈110 nm thick PBTTT film. After spin-coating, the 150 nm thick Au layer was deposited through a shadow mask at a pressure of approximately 10^{-7} Torr at a rate of 7 Å s^{-1} to form S and D electrodes with a channel length and width of 30 µm and 1 mm, respectively. The PBTTT films were thermally annealed at 180 °C for 20 min under vacuum conditions. During this process, depending on the type of OFET device, a pressure of 0.49 kPa was applied to the various samples either simultaneously or separately to complete the devices. To apply pressure to the prepared devices, a hydrophobic ODTS-modified SiO_2/Si wafer was placed over the devices, and then a weight was placed on top. The magnitude of the pressure was calculated considering the mass of the weight placed and the size of the SiO_2/Si wafer.

Characterizations: The morphologies of the active layers were visualized by ex situ tapping-mode atomic force microscopy (AFM; Digital Instruments MultiMode). The SiN_x cantilevers on the Si tips have nominal constants and frequency values of 42 and 320 kHz, respectively. The AFM data were analyzed using Nanoscope Analysis 1.40 software. To investigate the crystalline structure of the PBTTT layer, 2D GIXD analysis was performed at a wavelength of 1.02 Å. The analysis was performed at the Pohang Accelerator Laboratory (PAL), Korea. The electrical parameters of the PBTTT OFETs were determined at room temperature using a Hewlett–Packard HP4156A semiconductor parameter analyzer in the dark, under ambient environments. The fundamental performance parameters of the OFET devices were calculated in the saturation regime of the transfer characteristics. The C_i value of the 300 nm thick SiO_2 gate dielectrics was measured using a sandwich structure between the Au dots and was estimated as 10.8 nF cm^{-2} using an Agilent 4284 precision inductance–capacitance–resistance (LCR) meter. The computational details of the finite element analysis are provided in the Supplementary Information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Keywords

buried-contact OFET, contact resistance, deep trap, four-probe measurement, top-contact OFET, transfer line method

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